Binary Classification

CSC 272 - Machine Learning

Aum Hirpara and Altaf Shaik

Abstract*:* In this research, we use two datasets, the Titanic and the Breast Cancer Wisconsin (Diagnostic) dataset, to investigate the nuances of binary categorization. The main goal is to compare the effectiveness of four well-known machine learning algorithms: Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Perceptron. To guarantee the quality and usefulness of the datasets, the study starts with data pre-processing procedures. Next, the chosen algorithms are implemented. To maximize model performance, hyper parameter tweaking is carried out, and several studies are carried out to comprehend the influence of training dataset size and computing efficiency.

As can be seen from the findings, each method has advantages and disadvantages. Decision Trees work particularly well at collecting intricate patterns in both datasets. The significance of feature selection, interpretability of the model, and the impact of class imbalance on model performance are emphasized in the study. The project's insights can guide future investigations into binary categorization, opening the door to more advanced machine learning applications in practical settings. The results highlight the need of thorough assessment in obtaining precise forecasts and streamlining machine learning processes.

1. Introduction:

In the data-driven world of today, machine learning is essential to making wise judgements in a variety of fields. Binary classification is a core job in machine learning, where the goal is to classify examples into one of two unique classes. By putting several machine learning algorithms into practice and examining them, this research aims to address the difficulties and subtleties of binary categorization.

Two exciting binary classification problems are presented by the Titanic and the Breast Cancer Wisconsin (Diagnostic) datasets, which we shall analyses in this work. Using this approach, we want to evaluate and compare the performance of four popular machine learning algorithms: Decision Tree, K-Nearest Neighbors (KNN), Perceptron, and Logistic Regression. By detailing our procedures, findings, and observations, the study will shed light on how these algorithms function in different situations.

2. Datasets:

1. Breast Cancer Wisconsin (Diagnostic) Dataset

The area of medical diagnostics makes extensive use of the Breast Cancer Wisconsin (Diagnostic) dataset. It consists of 569 examples with 30 traits that represent different aspects of the cell nuclei seen in breast cancer biopsies. The binary target variable indicates if the tumor is benign (B) or malignant (M). This dataset's characteristics are numerical and come from digital photos of specimens with breast masses. These features include:

- **Radius** Mean: The average distance from the center to points on the perimeter of the tumor.
- **Texture** Mean: The standard deviation of the gray-scale values.

- **Perimeter** Mean: The average perimeter length of the tumor.
- **Area** Mean: The average area of the tumor.

This dataset is especially intriguing since it represents a practical use of machine learning in the medical field, where increasing the precision of tumor identification has the potential to save lives. It also makes it possible to investigate the significance of features, the interpretability of models, and the effects of different algorithms on a crucial healthcare problem.

2. **Titanic Dataset**

An historical collection of data regarding the people who travelled on the tragic Titanic ship is called the Titanic dataset. 887 occurrences are included, along with a number of other variables including ticket class, demographic information, and survival status (0 = Not Survived, 1 = Survived). Principal attributes consist of:

- **PassengerId**: Unique identifier for each passenger.
- **Pclass**: Class of the ticket (1st, 2nd, or 3rd).
- **Sex**: Gender of the passenger.
- **Age**: Age of the passenger at the time of the voyage.
- **Fare:** The fare paid for the ticket.

This dataset is interesting because it depicts a situation in real life where survival was influenced by a number of variables, including gender, age, and socioeconomic status. Many studies have been conducted on the Titanic dataset, which is a great example of a test case for machine learning algorithms because of its rich historical context and variety of variables.

Pre-processing: For both datasets, we conducted essential pre-processing steps to ensure the quality and usability of the data:

1. **Handling Missing Data:** To make the study easier, we decided to remove rows in the Titanic dataset that had missing values. This choice was taken with the understanding that more sophisticated imputation methods may be used in subsequent research in order to preserve data integrity. There were no missing values in the Breast Cancer dataset.
2. **Feature Scaling:** To make sure that every input variable contributed equally to the distance calculations in algorithms such as KNN, we standardized the features. This was especially crucial for distance-based algorithms since different scales might cause findings to be skewed.
3. **Data Encoding:** To make them compatible with machine learning techniques, categorical characteristics (such "Sex" in the Titanic dataset) were converted into numerical values.

Hypotheses: We made the following assumptions about the algorithms' performance based on the features of the two datasets:

Breast Cancer Dataset: Given the characteristics nature and logistic regression's linearity, we anticipated that this dataset would respond very well to the model. In addition, decision trees and KNN could work well because of their capacity to manage intricate interactions.

Titanic Dataset: Given their ability to capture the non-linear correlations included in the demographic data, tree-based techniques like Decision Trees and KNN were expected to perform well. Because of the non-linear nature of the interactions between the characteristics, Logistic Regression may not perform well on this dataset.

3. <u>Methods</u>: To evaluate the performance of the machine learning algorithms, we employed the following methods:

1. **Train-Test Split**: Using an 80-20 split, each dataset was separated into training and testing sets. In order to avoid overfitting, we were able to train the models on one portion of the data and assess their performance on a different fraction using this method.

2. **Model Selection**: We selected four algorithms for our analysis:

Decision Tree: A versatile algorithm that can model complex non-linear relationships by splitting the data into homogenous groups based on feature values.

K-Nearest Neighbors (KNN): A simple, instance-based learning algorithm that classifies instances based on the majority class of their nearest neighbors.

Perceptron: A type of linear classifier that updates its weights based on misclassifications during training.

Logistic Regression: A linear model used for binary classification that estimates probabilities using a logistic function.

<u>4. Hyperparameter Tuning</u>: For each algorithm, we performed hyperparameter tuning using GridSearchCV to identify the optimal settings. This process involved testing various hyperparameters and observing their effect on model performance. The following parameters were tuned:

Decision Tree: **max_depth:** Values ranging from 1 to 10.

KNN: **n_neighbors**: Values ranging from 1 to 20.

Perceptron: **max_iter**: Values ranging from 200 to 1000.

Logistic Regression: **C:** Values ranging from 10^-3 to 10^3.

Observations: The hyperparameter tuning results provided valuable insights into each algorithm's behavior:

1. **<u>Decision Tree</u>:** In both datasets, the training score increases significantly with depth, while the cross-validation score does not increase as sharply. This indicates overfitting, where the model learns the training data too well and performs poorly on unseen data.

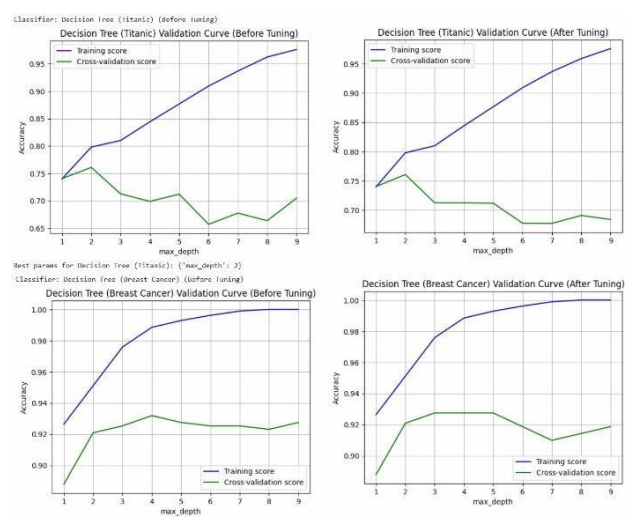

Figure 1: Decision Tree

After Tuning show improvements in cross-validation scores, suggesting that hyperparameter tuning (like adjusting the maximum depth of the tree) helps to balance model complexity and generalization. **Overall**, the model demonstrated substantial improvements in accuracy as max_depth increased, indicating that deeper trees captured more complexity. However, overfitting became apparent with excessively deep trees.

2. **<u>K-Nearest Neighbors</u>:** In both datasets, the training score decreases while the cross-validation score fluctuates. This suggests that as the model complexity changes with the number of neighbors, it may overfit the training data, especially at lower neighbor counts.

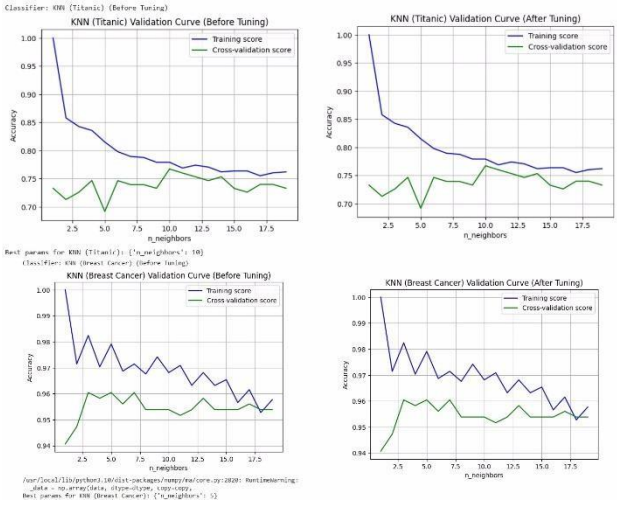

Figure 2: K-Nearest Neighbors

The graph illustrates hyperparameter tuning (adjusting the number of neighbors) can lead to better cross-validation performance, enhancing the model's ability to generalize. **Overall**, As the number of neighbors increased, the model's accuracy improved initially before stabilizing. This suggested that while more neighbors can provide a more generalized prediction, too many can dilute the decision boundary.

3. **<u>Perceptron</u>:** Observing both datasets data, the training score tends to remain stable as the maximum iterations increase, but the cross-validation score's fluctuations indicate variability in performance on unseen data.

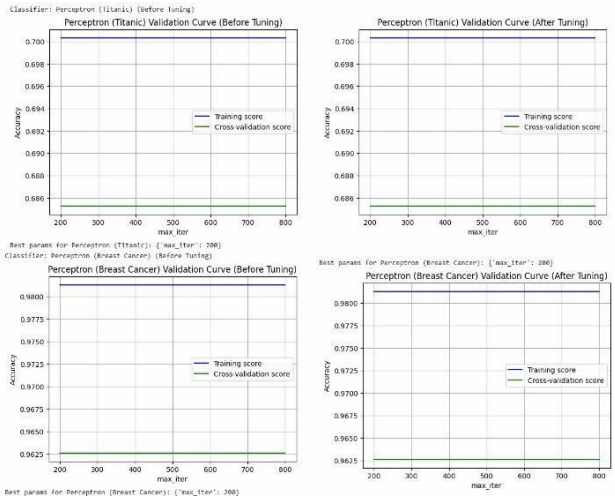

Figure 3: Perceptron

The tuning process helps to find the optimal number of maximum iterations that balances training and validation performance, leading to improved generalization. Monitoring both training and cross-validation scores is crucial for understanding the model's behavior and helps in making informed decisions on hyperparameter adjustments. **Overall**, the accuracy improved as we increased the number of iterations. However, it reached a plateau beyond a certain point, indicating diminishing returns on computational resources.

4. **<u>Logistic Regression:</u>** The initial graphs indicate potential overfitting if the training score is much higher than the cross-validation score. The goal is to find a balance where both scores are high and close together.

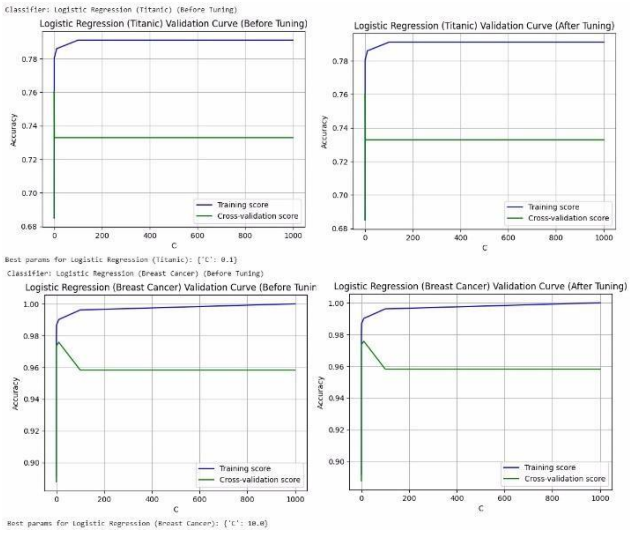

Figure 4: Logistic Regression

The C parameter is crucial for controlling the degree of regularization. Finding an optimal C helps to reduce overfitting and improve the model's ability to generalize. The divergence between training and cross-validation scores serves as an important diagnostic tool. A large gap suggests that the model is too complex for the data, while a smaller gap with high scores indicates a well-tuned model. **Overall**, the performance fluctuated with the C parameter. A lower value of C resulted in more regularization, which improved generalization but may lead to underfitting. Conversely, high values could cause overfitting.

5. <u>Other Analyses</u>:

- **Training Dataset Size Effects:** To further explore how the size of the training dataset impacted model performance, we conducted experiments with different proportions of training data (e.g., 60-40, 70-30). This analysis revealed interesting trends:
  - **Breast Cancer Dataset**: As the training set size increased, model accuracy consistently improved, demonstrating that more data provides better learning opportunities.
  - **Titanic Dataset**: The results were mixed; while larger training sets generally improved performance, specific models like KNN experienced diminishing returns beyond a certain size due to their reliance on the local neighborhood.

- **Training/Testing Timing Analysis:** We also performed a timing analysis to compare the computational efficiency of each algorithm. The results indicated that:

The plots are shown below for before tuning and after tuning:

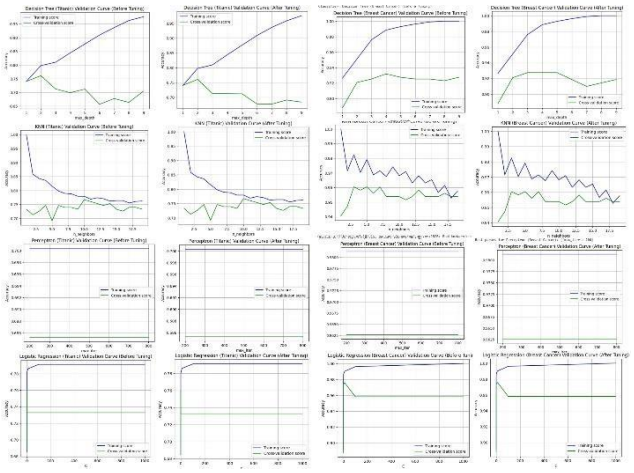

Figure 5: Plots Before & After Tuning

- **Decision Trees** fell somewhere in between, with reasonable training times but potential for longer inference times as the tree depth increased.
- **KNN** exhibited significantly longer training times as the dataset grew, particularly because it requires distance calculations for all instances.
- **Perceptron** and **Logistic Regression** were the fastest to train, making them suitable for large datasets.

6. **Results:** After optimizing the hyperparameters, we summarized the final results of each algorithm in a table:

Classifier Model	Accuracy Before Tuning	Accuracy After Tuning	Training Time Before	Training Time After
Decision Tree (Titanic)	0.7027027027027027	0.8108108108108109	0.0026369094848632812	0.00866246223449707
Decision Tree (Breast Cancer)	0.9473684210526315	0.9385964912280702	0.038393259048461914	0.01912689208984375
KNN (Titanic)	0.7297297297297297	0.7567567567567568	0.0064220428466796875	0.002529144287109375
KNN (Breast Cancer)	0.9473684210526315	0.9473684210526315	0.002472400665283203	0.0024683475494384766
Perceptron (Titanic)	0.5405405405405406	0.5405405405405406	0.0033004283905029297	0.002863645553588867
Perceptron (Breast Cancer)	0.9736842105263158	0.9736842105263158	0.005301952362060547	0.004266023635864258
Logistic Regression (Titanic)	0.7297297297297297	0.7297297297297297	0.0053234100341796875	0.004852294921875
Logistic Regression (Breast Cancer)	0.9736842105263158	0.9736842105263158	0.02006220817565918	0.03902435302734375

7. Discussion

Performance Comparison

Based on the results, we observed that:

- The Decision Tree algorithm excelled in both datasets, showcasing its flexibility in modeling complex patterns.
- KNN delivered competitive accuracy, particularly in the Breast Cancer dataset, but suffered in the Titanic dataset due to the challenges of local density estimation.
- Perceptron performed decently but was less effective in comparison to other algorithms, particularly in datasets with non-linear relationships.
- Logistic Regression showed reliable performance in the Breast Cancer dataset, benefiting from its linear nature.

Insights and Interpretations

1. **Feature Importance**: In the Breast Cancer dataset, analyzing feature importance through the Decision Tree model highlighted the most influential features for classifying tumors. For instance, features like radius and texture significantly contributed to accurate predictions.
2. **Class Imbalance**: Both datasets exhibited some level of class imbalance, particularly the Titanic dataset. Future work should include some techniques to balance the classes, improving model performance.
3. **Model Interpretability**: Decision Trees provided the most intuitive and interpretable results, allowing stakeholders to understand the decision-making process. In contrast, models like KNN and Logistic Regression were less interpretable.
4. **Limitations**: This analysis had certain limitations, such as the simplistic approach to missing data in the Titanic dataset and the lack of consideration for ensemble methods that could potentially enhance performance.

8. **Conclusion:** To sum up, this study offered a thorough investigation of binary classification by utilizing a variety of algorithms on the Titanic and Breast Cancer Wisconsin datasets. We learnt a great deal about the advantages and disadvantages of Decision Tree, KNN, Perceptron, and Logistic Regression by analyzing their performance. In order to get accurate predictions, the study reaffirmed the need of thorough data pre-processing, hyperparameter tweaking, and model validation.
The knowledge acquired from this effort will provide the groundwork for next investigations and advancements in the field of binary classification as machine learning continues to develop. Our aim is to make a more significant contribution to the current discussion about efficient classification methods and their practical applications.

9.References:

- https://www.kaggle.com/competitions/titanic
- https://archive.ics.uci.edu/dataset/14/breast+cancer